DEPARTMENT OF
COMPUTER SCIENCE

NOVA SCHOOL OF
SCIENCE & TECHNOLOGY

CAROLINA VIEIRA SIMONET

BSc in Computer Science and Engineering

AI-POWERED REQUIREMENTS ENGINEERING

A NEW APPROACH

MASTER IN COMPUTER SCIENCE AND ENGINEERING

NOVA University Lisbon
Draft: December 29, 2024

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Abstract

In software development, requirements engineering plays a critical role in ensuring that software products meet stakeholder expectations and standards. However, the manual writing and thinking of requirements gathering, refinement, and validation introduces the risk of misunderstanding, incomplete requirements, and communication breakdowns between stakeholders and development teams.

Although AI techniques have been explored in various stages of software development, their application in automating and improving requirements engineering is still a new field of study. This thesis aims to explore the use of Generative AI (GAI) to automate the identification, clarification, and specification of requirements, ultimately improving the efficiency and accuracy of the RE process.

Concerning its contents, the abstracts should not exceed one page and may answer the following questions (it is essential to adapt to the usual practices of your scientific area):

1. What is the problem?
2. Why is this problem interesting/challenging?
3. What is the proposed approach/solution/contribution?
4. What results (implications/consequences) from the solution?

Keywords: One keyword, Another keyword, Yet another keyword, One keyword more, The last keyword

Resumo

Palavras-chave: Primeira palavra-chave, Outra palavra-chave, Mais uma palavra-chave, A última palavra-chave

Contents

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Introduction

1.1 Motivation and Context

This first Chapter introduces the template and how it is organized. In Chapter **??** you can find some specific instructions on how to use this template. Chapter **??** shows some examples and give some hints on how to write your text. Please read these next Chapters carefully.

Teste caso queira usar

1.2 Problem Definition

Problem definition

1.3 Objectives

Thesis objectives

1.4 Proposed Solution

Proposed Solution

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Background

2.1 Introduction

This Chapter describes how to use the template. It is assumed that you have a working of LaTeX, either local (in your own computer) or remote, and that you were able to generate a PDF for the default c

2.2 Quick Start

2.2.1 With a Local LaTeX Installation

Follow these steps to get started with a local LaTeX installation:

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Related Work

3.1 Document Structure

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Work Plan and Proposal

Need a ganttChart